

Brain functional magnetic resonance imaging response to glucose and fructose infusions in humans.

AIMS: In animals, intracerebroventricular glucose and fructose have opposing effects on appetite and weight regulation. In humans, functional brain magnetic resonance imaging (fMRI) studies during glucose ingestion or infusion have demonstrated suppression of hypothalamic signalling, but no studies have compared the effects of glucose and fructose. We therefore sought to determine if the brain response differed to glucose vs. fructose in humans independently of the ingestive process.

METHODS: Nine healthy, normal weight subjects underwent blood oxygenation level dependent (BOLD) fMRI measurements during either intravenous (IV) glucose (0.3 mg/kg), fructose (0.3 mg/kg) or saline, administered over 2 min in a randomized, double-blind, crossover study. Blood was sampled every 5 min during a baseline period and following infusion for 60 min in total for glucose, fructose, lactate and insulin levels. **RESULTS:** No significant brain BOLD signal changes were detected in response to IV saline. BOLD signal in the cortical control areas increased during glucose infusion ($p = 0.002$), corresponding with increased plasma glucose and insulin levels. In contrast, BOLD signal decreased in the cortical control areas during fructose infusion ($p = 0.006$), corresponding with increases of plasma fructose and lactate. Neither glucose nor fructose infusions significantly altered BOLD signal in the hypothalamus. **CONCLUSION:** In normal weight humans, cortical responses as assessed by BOLD fMRI to infused glucose are opposite to those of fructose. Differential brain responses to these sugars and their metabolites may provide insight into the neurologic basis for dysregulation of food intake during high dietary fructose intake. © 2011 Blackwell Publishing Ltd.